

Hyperparameter Relationship Analysis in Evolutionary Poker AI Training Tournament Synthesis v2

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Abstract

This report presents an expanded empirical synthesis of hyperparameter relationships governing evolutionary self-play training in Texas Hold'em poker. Compared to prior analyses, this study incorporates 33 unique configurations and 4,492 tournament matches spanning multiple evaluation batches. We validate previously identified scaling relationships and introduce refined insights on population size, matchup diversity, mutation strength, and evaluation budget. Results indicate that smaller populations benefit from higher opponent diversity and lower mutation noise, while excessive matchups or evaluation depth can degrade learning stability. The strongest configuration achieves an 81.2% tournament win rate using a small population with high matchup diversity and low mutation sigma. These findings provide practical scaling heuristics for compute-efficient evolutionary training in imperfect-information games.

1 Overview and Dataset Expansion

This tournament synthesis expands upon earlier hyperparameter analyses by integrating Batch 1, Batch 2, Batch 1+2 championship results, and additional validation batches. In total, the dataset includes:

- 33 unique evolutionary training configurations
- 4,492 tournament matches
- Multiple independent tournament batches for cross-validation
- At least 6 tournament evaluations per configuration
- A minimum of 108 games per configuration

The best performing configuration in the updated dataset is:

`p12_m8_h500_s0.08_g200`

which achieved an 81.2% win rate, demonstrating that small populations combined with high opponent diversity and low mutation noise can outperform larger population regimes under specific conditions.

2 Population Size vs. Matchups per Agent

We confirm a robust inverse relationship between population size and the required number of opponent matchups per agent. Smaller populations require proportionally more matchups to compensate for reduced intrinsic strategic diversity.

Population	Matchups	Ratio (m/p)	Win Rate	Performance Tier
12	8	0.67	81.2%	Best (Batch 2)
12	6	0.50	61–68%	Good
20	6	0.30	71.3%	Good
40	8	0.20	78.7%	Best (Batch 1)
40	6	0.15	73.1%	Good

Table 1: Scaling relationship between population size and matchup count.

Design Implications:

- Large populations ($p \geq 40$) perform optimally near $m = 8$ (approximately 20% matchup ratio)
- Small populations ($p = 12$ – 20) require matchup ratios above 50% for stable fitness estimation
- Excessive matchups in small populations (e.g., $p12 + m10$) introduce noisy training signals and degrade performance

3 Matchups vs. Hands per Matchup

A consistent tradeoff emerges between opponent diversity and evaluation depth. At fixed evaluation budgets, increasing opponent variety yields greater performance gains than increasing hands per matchup.

Matchups	Hands	Total Evaluations	Win Rate	Verdict
8	375	3,000	78.7%	Best (Batch 1)
8	500	4,000	81.2%	Best (Batch 2)
6	500	3,000	68–73%	Good
6	750	4,500	61–67%	Slower, no gain
6	375	2,250	59–64%	Borderline
10	375	3,750	16–40%	Excessive diversity noise

Table 2: Opponent diversity provides stronger learning signal than deeper evaluation.

Performance saturates beyond approximately 3,000–4,500 total hands per agent, indicating diminishing returns from increasing evaluation depth alone.

4 Population Size vs. Mutation Sigma

Mutation noise exhibits a strong interaction with population size. While mutation promotes exploration, excessive noise destabilizes training, particularly in small populations.

Population	Sigma (σ)	Win Rate	Stability Assessment
12	0.08	81.2%	Optimal
12	0.10	60–64%	Acceptable
12	0.12	44–54%	Borderline
12	0.15	22–68%	Unstable
20	0.15	71.3%	Acceptable
40	0.10	78.7%	Optimal
40	0.15	73.1%	Slightly Excessive

Table 3: Mutation strength sensitivity across population sizes.

Across tested regimes, mutation stability follows the empirical scaling law:

$$\sigma \approx \frac{0.5}{\sqrt{p}}$$

indicating mutation magnitude should decrease as population diversity increases.

5 Proven High-Performance Configuration Families

5.1 Small Population, High Matchups, Low Sigma

Population: 12
Matchups: 8
Hands: 500
Sigma: 0.08
Generations: 200

Win Rate: 81.2%

Strength: Maximum performance under compute-efficient regime

5.2 Large Population, Standard Sigma

Population: 40
Matchups: 8
Hands: 375
Sigma: 0.10
Generations: 50

Win Rate: 78.7%

Strength: Fast training, strong generalization, rapid iteration

5.3 High Diversity, High Sigma

Population: 40
Matchups: 6
Hands: 500
Sigma: 0.15
Generations: 200

Win Rate: 73.1%

Strength: Exploration-heavy long-horizon training

6 Training Efficiency Tradeoffs

Configuration	Training Time	Win Rate	Intended Use Case
p12_m8_h500_s0.08_g200	98 min	81.2%	Maximum performance
p40_m8_h375_s0.10_g50	29 min	78.7%	Rapid experimentation
p40_m10_h500_s0.08_g50	44 min	Predicted strong	Balanced regime
p60_m12_h375_s0.06_g50	39 min	Predicted strong	High diversity

Table 4: Performance versus training time tradeoffs.

7 Identified Anti-Patterns

The following configuration regimes consistently degrade training performance:

- Small populations with high mutation noise ($p12 + \sigma = 0.15$)
- Insufficient matchups in large populations ($p40 + m6 + g50$)
- Excessively large evaluation budgets ($h = 1000$)
- Excessive matchup counts in small populations ($p12 + m10$)
- Low opponent diversity producing noisy fitness estimates

8 Methodology Summary

- Batch 1: 1,080 matches
- Batch 2: 1,872 matches
- Batch 1+2 Championship: 544 matches
- Additional validation batches: 996 matches
- Round-robin tournament format (10,000 hands per matchup)
- Minimum 6 tournament runs per configuration
- Minimum 108 games per configuration

9 Limitations

- Population sizes above 40 remain extrapolated rather than fully tested
- Sigma tested only at discrete values (0.08, 0.10, 0.12, 0.15)
- Generation counts evaluated primarily at 50 and 200
- Results currently specific to heads-up Texas Hold'em

10 Future Directions

- Validate predicted $p50$ and $p60$ scaling regimes
- Fine-grained sigma tuning (0.07, 0.09, 0.11)
- Matchup optimization studies ($m = 9, 10, 11$)
- Generation length refinement ($g = 30, 40, 100$)
- Cross-validation across top-performing configuration families

11 Conclusion

This expanded tournament synthesis provides strong evidence that evolutionary poker training follows reproducible hyperparameter scaling laws. Effective training depends more on opponent diversity, mutation stability, and evaluation budget balance than brute-force population scaling. These findings support compute-efficient evolutionary learning design in imperfect-information environments.

12 References and Reports

Tournament data and detailed logs are available in:

- `tournament_reports/overall_reports/Batch1_Report/`
- `tournament_reports/overall_reports/Batch2_Report/`
- `tournament_reports/overall_reports/Batch1and2_Report/`
- `tournament_reports/tournament_results_analysis.txt`